

LM Physics of Climate

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1 Some atmospheric fundamentals

1.1 A planet's energy budget: equilibrium temperature

A planet's energy budget is fundamental to determining its climate. Where does the Earth get its energy from? A small part of it is heat emanating from the interior (a combination of radioactive decay and the leftover heat from the formation of the Earth), but most of the energy budget comes from the Sun.

We can characterise the energy budget that a planet receives from its host star (in our case, the Sun) by its *equilibrium temperature*, T_{eq} . If we assume the atmosphere to be emitting blackbody radiation at a single, homogenous temperature, T_{eq} is the temperature it would need to be at to be in equilibrium with the incoming stellar radiation.

Recall that the luminosity (or energy emitted per unit time) of a blackbody radiator with surface area A and temperature T is $A\sigma T^4$, where σ is the Stefan-Boltzmann constant. Assuming both the star and planet to emit and absorb radiation as blackbodies, the stellar luminosity received at the planet is:

$$L_{in} = 4\pi R_{\star}^2 \sigma T_{\star}^4 \frac{\pi R_p^2}{4\pi a^2},$$

where $R_{\star}$ and R_p are the radii of the star and planet, respectively, and $T_{\star}$ is the star's effective temperature.

Similarly, the outgoing luminosity from the planet is:

$$L_{out} = 4\pi R_p^2 \sigma T_p^4,$$

where T_p is the temperature of the planet.

In equilibrium, $L_{in} = L_{out}$, and solving for the planetary temperature gives us

$$T_{eq} = \sqrt{\frac{R_{\star}}{2a}} T_{\star}. \quad (1.1)$$

For many planets in the Solar System (and some beyond it), a fraction of the incoming solar radiation is reflected or scattered, for example by clouds. This fraction is known as the Bond albedo, A_B , and is equal to roughly 0.3 for the Earth. The Bond albedo modifies the luminosity received by a planet:

$$L_{in} = 4\pi R_{\star}^2 \sigma T_{\star}^4 (1 - A_B) \frac{\pi R_p^2}{4\pi a^2},$$

and hence the equilibrium temperature becomes

$$T_{eq} = (1 - A_B)^{1/4} \sqrt{\frac{R_{\star}}{2a}} T_{\star}. \quad (1.2)$$

Let's see what T_{eq} is for some familiar planets. The Sun has $R_{\star} = 6.95 \times 10^8$ m and $T_{\star} = 5777$ K.

- For the Earth, $a = 1 \text{ au} = 1.5 \times 10^{11} \text{ m}$ and $A_B = 0.3$, which results in $T_{eq} = 255 \text{ K}$.
- For Venus, $a = 0.72 \text{ au}$ and $A_B \approx 0.75$, resulting in $T_{eq} = 232 \text{ K}$.

This may be surprising, since Venus' surface is famously much hotter than Earth's (740 K vs 288 K¹). However, Venus' highly reflective, sulphuric acid clouds scatter so much incident irradiation that its atmosphere actually receives less energy from the Sun compared to the Earth. Instead, it is Venus' thick atmosphere which makes its surface so warm.

1.2 A simple climate model: adding in an atmosphere

Let's consider a scenario somewhat similar to the Earth, where the atmosphere is transparent at visible wavelengths and partially opaque in the infrared. The incoming solar radiation goes straight through the atmosphere and heats up the surface, whose outward radiation peaks in the infrared and is intercepted by the atmosphere.

Unlike our simplified planet and star, the atmosphere is partly transparent and therefore does not emit as a blackbody. Instead, its emission can be described as blackbody emission modulated by an efficiency factor, ε_{IR} , which can take values between 0 and 1. $\varepsilon_{IR} = 0$ corresponds to no emission, while $\varepsilon_{IR} = 1$ corresponds to perfect, blackbody-like emission.

If we consider the atmosphere as a 'slab' with surface area A at both the top and bottom, its outgoing luminosity becomes $2A\varepsilon_{IR}\sigma T_a^4$, where T_a is the temperature of the atmosphere. According to Kirchhoff's law of thermal radiation, a material's absorptivity is equal to its emissivity. Hence, the luminosity received by the atmosphere from the surface is $A\varepsilon_{IR}\sigma T_s^4$, where T_s is the surface temperature. Equating the two, we find that

$$T_a = 2^{-1/4}T_s, \quad (1.3)$$

which is more commonly known as the *skin temperature*.

We can separately consider the energy balance of the surface. The luminosity per unit area 'in' from the stellar irradiation is σT_{eq}^4 , from the atmosphere it is $\varepsilon_{IR}\sigma T_a^4$, and the luminosity per unit area 'out' is σT_s^4 . Balancing the 'in' and 'out' terms gives us

$$\sigma T_{eq}^4 + \varepsilon_{IR}\sigma T_a^4 = \sigma T_s^4. \quad (1.4)$$

Combining Equations 1.3 and 1.4, we get

$$T_s = \frac{T_{eq}}{(1 - \varepsilon_{IR}/2)^{1/4}}. \quad (1.5)$$

When $\varepsilon_{IR} = 0$, it is as if the atmosphere does not exist and $T_s = T_{eq}$. If, on the other hand, $\varepsilon_{IR} = 1$, the atmosphere is completely opaque and $T_s = 2^{1/4}T_{eq}$. For the Earth, this temperature would be 303 K, i.e. warmer than the current surface temperature. This makes sense, since in reality Earth's atmosphere is not opaque at all wavelengths of light.

In the case of Venus, $T_s(\varepsilon_{IR} = 1) = 276 \text{ K}$, which is much colder than its true surface temperature. This simple model does not work for Venus because its atmosphere is not transparent in the optical, as we assumed above.

Of course, even Earth's atmosphere is much more complex than the model described above, as we will discover throughout this course. Figure 1 shows the Earth's average temperature profile as a function of altitude, and it is worth pointing out some features

¹Earth value taken from 2024, <https://wmo.int/news/media-centre/wmo-confirms-2024-warmest-year-record-about-155degc-above-pre-industrial-level>

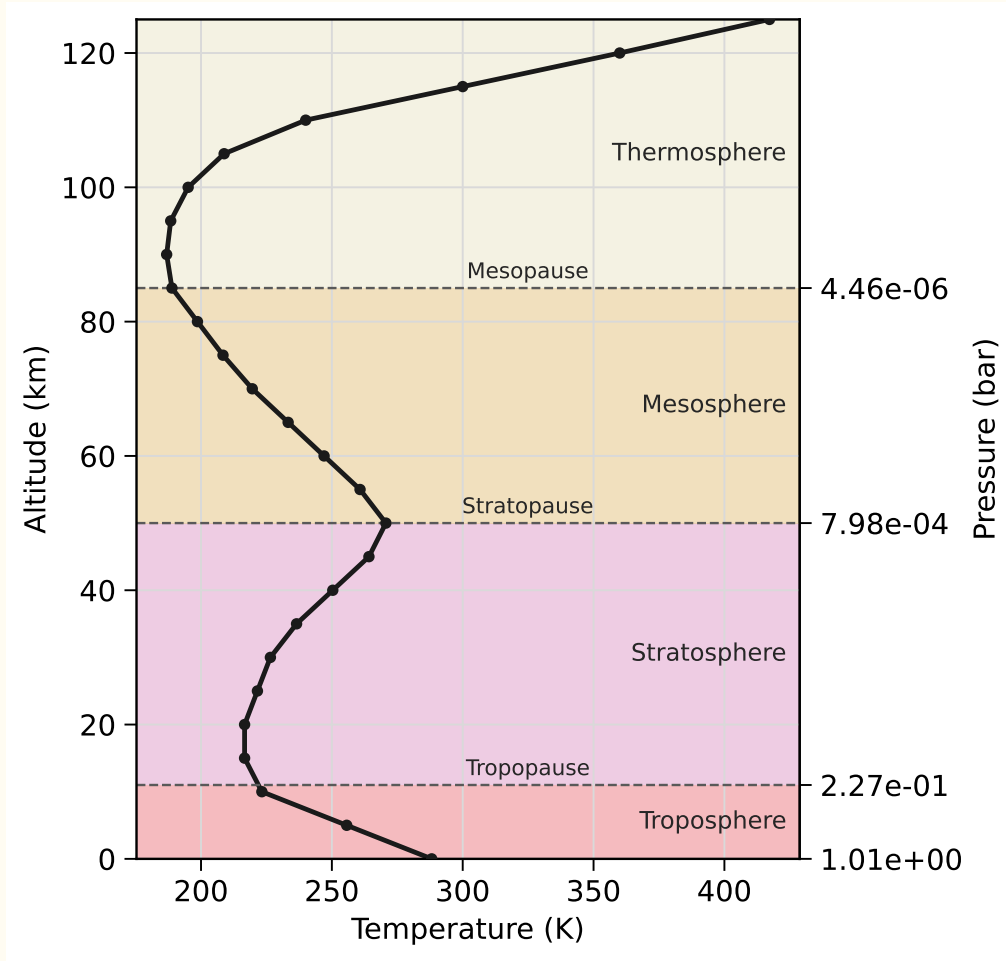


Figure 1: An average vertical temperature profile of the Earth's atmosphere. Data are from the U.S. Standard Atmosphere, 1976 (Published here: https://www.ngdc.noaa.gov/stp/space-weather/online-publications/miscellaneous/us-standard-atmosphere-1976/us-standard-atmosphere_st76-1562_noaa.pdf).

which we will discuss in more detail later on. Notice that, while temperature initially decreases as you go up from the surface, it later *increases* again. These types of regions are generally called *thermal inversions*, and all of the solar system planets with atmospheres are known to have them.

1.3 Introducing pressure: hydrostatic equilibrium

How does the pressure in an atmosphere vary with altitude? For the case of a stable, non-escaping atmosphere, we can work this out by balancing the forces on a parcel of gas.

Consider the cylinder of gas in Figure 2, with cross-sectional surface area A , density ρ and height δz . The pressure at the base of the parcel is P , while at the top the pressure is $P + \delta P$. In a hydrostatic atmosphere, this imbalance of pressure forces is compensated by the gravitational force. Note that in an atmosphere, δP is negative.

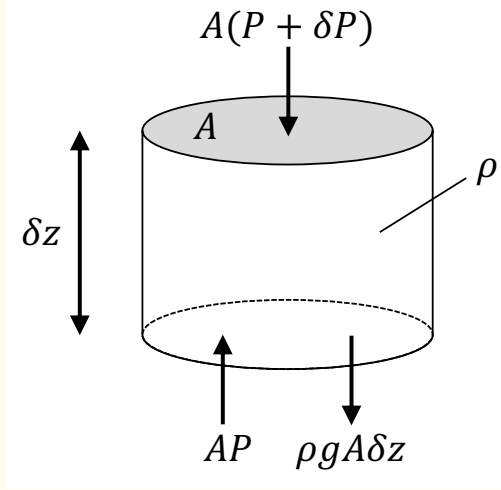


Figure 2: The pressure and gravitational forces acting on a parcel of gas with density ρ , height δz , cross-sectional surface area A and gravitational acceleration g .

Balancing the upwards and downwards forces on the parcel of gas, we obtain

$$\frac{dP}{dz} = -\rho g. \quad (1.6)$$

To solve this equation, we can use the ideal gas law to write ρ in terms of P and T : $P = \frac{k_B}{\mu} \rho T$, where ρ is the mass density of the gas and μ is its mean molecular weight (e.g. for pure H_2 gas $\mu = 2 \text{ u} = 3.32 \times 10^{-27} \text{ kg}$). This substitution gives:

$$\frac{d \ln P}{dz} = -\frac{\mu g}{k_B T}. \quad (1.7)$$

In the simple case of an isothermal atmosphere, the solution to Equation 1.7 is

$$P = P_0 e^{-z/H_{sc}},$$

where the scale height of the atmosphere,

$$H_{sc} = \frac{k_B T}{\mu g}. \quad (1.8)$$

The scale height is the distance over which pressure decreases by a factor of e , and gives a sense of the vertical extent of an atmosphere. For example, an atmosphere which is hotter, has a lower mean molecular weight and/or a lower gravity has a ‘puffier’ atmosphere.

We can estimate H_{sc} for the Earth using its equilibrium temperature, $T_{eq} = 255 \text{ K}$, the mean molecular weight of dry air, $\mu = 28.97 \text{ u}$, and $g = 9.81 \text{ m/s}^2$. This gives us $H_{sc} = 7.46 \text{ km}$, or close to the height of Mount Everest (8.85 km).